Neural Computational Fluid Dynamics Under SCX Audit:

Finite Element Aerodynamics with Certified Quality Guarantees

SCX

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Abstract

We apply the SCX quality audit framework to neural computational fluid dynamics (CFD) for finite element aerodynamics. While traditional CFD employs multiple physically grounded "experts"—turbulence models (k-ε, k-ω SST, LES, DES), mesh convergence hierarchies, and distinct numerical discretizations—neural CFD solvers (physics-informed neural networks, Fourier neural operators, mesh graph networks, learned turbulence closures) lack formal quality guarantees. We formalize aerodynamic simulation as state-conditioned prediction over flow regimes, and prove three theorems: (1) an SCX noise detection bound that limits the probability of missing systematic errors when M solvers (traditional plus neural) are applied to the same geometry; (2) an unidentifiability theorem establishing that when a neural solver disagrees with wind tunnel data, the error source cannot be attributed without declared assumptions; and (3) a Situs encoding guarantee that relates geometric discretization fidelity in airfoil/wing representations to aerodynamic prediction error through Lipschitz continuity. We define the Cercis score for aerodynamic simulation, combining benchmark accuracy (ONERA M6, NASA CRM, NACA airfoils) with flow regime coverage novelty, and we rank published neural CFD methods. A multi-expert architecture with the Yajie consensus mechanism is proposed, using wind tunnel data as ground-truth anchor. We contribute an explicit assumption map for neural CFD and an experimental protocol covering standard benchmarks, turbulence model variations, mesh convergence studies, and neural architecture ablations. All claims are explicitly labeled with rigor level and assumption dependencies.
Keywords: SCX audit, neural CFD, finite element aerodynamics, turbulence modeling, aerodynamics, turbulence models, quality certification, multi-expert consensus

1 Introduction

Computational fluid dynamics (CFD) is the backbone of modern aerodynamic design, with finite element and finite volume methods enabling accurate simulation of flows around aircraft wings, turbomachinery blades, and automotive bodies. The governing equations—the Navier-Stokes equations—are well understood, but their direct numerical solution at industrially relevant Reynolds numbers remains computationally prohibitive. This has led to a rich ecosystem of modeling approximations, each constituting a distinct "expert" in the SCX sense.

1.1 Traditional CFD: A Natural Multi-Expert Landscape

Traditional CFD naturally provides multiple experts through independent modeling choices:

(i) **Turbulence models:** The Reynolds-Averaged Navier-Stokes (RANS) framework produces distinct closure models: k-ε [?], k-ω SST [?], Spalart-Allmaras [?], each making different assumptions about turbulent viscosity.

(ii) **Scale-resolving approaches:** Large Eddy Simulation (LES) [**?**] and Detached Eddy Simulation (DES) [**?**] resolve different fractions of the turbulent spectrum.

(iii) **Discretization schemes:** Finite volume vs. finite element vs. discontinuous Galerkin methods, each with distinct convergence properties and numerical dissipation characteristics.

(iv) **Mesh resolutions:** Grid convergence studies produce a family of solutions that can be treated as correlated experts with known Richardson extrapolation error estimates.

(v) **Experimental anchors:** Wind tunnel measurements [**? ?**] provide ground-truth data at specific Reynolds and Mach numbers, albeit with their own uncertainty.

These experts differ in computational cost, physical fidelity, and domain of applicability, but they share a common governing equation structure. The SCX insight is that this diversity is a *resource* for quality auditing, not a problem to be eliminated.

1.2 Neural CFD: Speed Without Guarantees

Recent years have seen a proliferation of neural approaches to CFD, each claiming dramatic speed advantages:

(i) **Physics-Informed Neural Networks (PINNs)** [**?**]: Solve PDEs by embedding governing equations into neural network loss functions. Applied to incompressible Navier-Stokes [**?**], but struggle with high Reynolds number turbulence.

(ii) **Fourier Neural Operators (FNOs)** [**?**]: Learn mappings between function spaces in Fourier domain. Applied to aerodynamics by **?**], showing speedups of 10^3–10^4 over traditional solvers for specific geometries.

(iii) **MeshGraphNets (GNN solvers)** [**?**]: Encode mesh-based simulations as graph neural networks. **?**] applied similar architectures to airfoil flow prediction.

(iv) **Learned turbulence closures**: Neural networks replace or augment traditional RANS turbulence models [**? ?**], learning corrections from high-fidelity LES or DNS data.

1.3 The Quality Gap

[A0] (Motivating Observation): No published neural CFD method provides a formal quality guarantee—i.e., a verified bound on prediction error for previously unseen geometries and flow conditions. The following specific concerns motivate the SCX audit:

(i) **Training data contamination:** Neural operators trained on aerodynamic databases may "memorize" specific geometries or flow features, giving misleadingly low errors on test splits drawn from the same distribution.

(ii) **Mesh dependency:** Neural solvers inherit biases from the resolution and topology of training meshes; prediction quality on meshes of substantially different density is unknown.

(iii) **Turbulence model mismatch:** Learned closures trained on one turbulence regime (e.g., attached boundary layers) may fail silently in another (e.g., separated flows).

(iv) **Architectural extrapolation:** Neural networks lack the asymptotic consistency guarantees that underlie traditional CFD methods (e.g., the Lax equivalence theorem).

1.4 Contributions

This paper provides the SCX audit framework for neural CFD:

(i) **Formal problem formulation** (§2): Aerodynamic simulation as state-conditioned prediction with explicit flow regime states.

(ii) **Noise detection theorem** (§3): Bounds on the probability of missing systematic neural CFD errors when audited by multiple traditional and neural experts.

(iii) **Error source unidentifiability theorem** (§4): Proof that without explicit assumption declaration, the source of neural CFD errors is fundamentally unidentifiable.

(iv) **Situs encoding guarantee** (§5): A Lipschitz-based bound relating geometric encoding fidelity to aerodynamic prediction error.

(v) **Cercis score** (§6): Quantitative ranking of existing neural CFD methods on standard benchmarks.

(vi) **Multi-expert architecture** (§7): Yajie consensus mechanism combining traditional and neural experts with wind tunnel anchoring.

(vii) **Experimental protocol** (§8): Reproducible benchmark suite with explicit assumption tracking.

We emphasize that this paper makes no "revolutionary" claims about neural CFD. Our contribution is a *quality audit framework* that enables rigorous comparison and certification of existing and future methods. **[L0]** The framework is descriptive and diagnostic; it does not prescribe how to improve neural CFD accuracy.

2 Problem Formulation: Aerodynamic Simulation as State-Conditioned Prediction

2.1 The Forward Aerodynamic Problem

Let $\Omega \subset \mathbb{R}^d$ ($d = 2$ for airfoils, $d = 3$ for wings) be the computational domain surrounding an aerodynamic body with boundary Γ_w (the wing/airfoil surface). The steady-state Reynolds-Averaged Navier-Stokes (RANS) equations are:

$$\begin{aligned} \nabla \cdot (\rho \mathbf{u}) &= 0 \\ \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) &= -\nabla p + \nabla \cdot \left[(\mu + \mu_t)(\nabla \mathbf{u} + \nabla \mathbf{u}^T)\right] \\ \nabla \cdot (\rho \mathbf{u} h) &= \nabla \cdot [(\lambda + \lambda_t)\nabla T] \end{aligned} \tag{1}$$

where ρ is density, $\mathbf{u}$ is velocity, p is pressure, μ is molecular viscosity, μ_t is turbulent (eddy) viscosity, h is enthalpy, λ is thermal conductivity, and λ_t is turbulent thermal conductivity. The turbulence model provides the closure for μ_t.

2.2 State-Conditioned Prediction Formulation

Definition 2.1 (Flow Regime State)**.** A *flow regime state* $s \in \mathcal{S}$ is a tuple:

$$s = (M_\infty, Re, \alpha, \Gamma, \tau) \in \mathcal{S} \tag{2}$$

where:
- M_∞: free-stream Mach number

- Re: Reynolds number based on chord length
- α: angle of attack
- $\Gamma \subset \mathbb{R}^d$: geometric description of the aerodynamic body (airfoil coordinates or wing surface mesh)
- $\tau \in \{\text{laminar}, \text{transitional}, \text{fully turbulent}, \text{separated}\}$: flow topology type

Definition 2.2 (Expert Solver)**.** An *expert solver* $f_i : \mathcal{S} \to \mathcal{O}$ maps a flow regime state s to an observable output:

$$f_i(s) = \big(C_L(s), C_D(s), C_M(s), \mathbf{C}_p(s; \Gamma_w)\big) \in \mathcal{O} \tag{3}$$

where C_L is lift coefficient, C_D is drag coefficient, C_M is moment coefficient, and $\mathbf{C}_p(s; \Gamma_w)$ is the pressure coefficient distribution over the aerodynamic surface.

[A1] (Deterministic Experts): Each expert f_i is a deterministic function of the flow regime state s, though different experts may produce different outputs for the same s. This abstracts away numerical noise below machine precision; for cases where numerical noise is non-negligible (e.g., under-resolved LES), we treat multiple realizations as distinct experts.

[A2] (Observable Domain): The output space $\mathcal{O}$ is a compact metric space with metric $d_{\mathcal{O}}$, e.g., the L^2 norm of pressure coefficient differences plus absolute lift/drag/moment differences:

$$d_{\mathcal{O}}(\mathbf{o}_1, \mathbf{o}_2) = \frac{|C_{L,1} - C_{L,2}|}{\max(|C_{L,1}|, 1)} + \frac{|C_{D,1} - C_{D,2}|}{\max(|C_{D,1}|, 1)} + \|\mathbf{C}_{p,1} - \mathbf{C}_{p,2}\|_{L^2(\Gamma_w)} \tag{4}$$

2.3 Expert Categories

We distinguish three categories of experts:

(i) **Traditional CFD experts** $\mathcal{F}_{\text{trad}} = \{f_1^{\text{RANS}}, f_2^{\text{LES}}, f_3^{\text{DES}}, \dots\}$: Physically motivated solvers with known assumptions and convergence properties.

(ii) **Neural CFD experts** $\mathcal{F}_{\text{NN}} = \{f_1^{\text{PINN}}, f_2^{\text{FNO}}, f_3^{\text{GNN}}, f_4^{\text{NN-closure}}, \dots\}$: Data-driven solvers with learned parameters.

(iii) **Ground-truth anchors** $\mathcal{F}_{\text{GT}} = \{f_1^{\text{exp}}\}$: Wind tunnel or flight test measurements at specific states.

[A3] (Ground-Truth Anchor): The wind tunnel expert $f^* = f_1^{\text{exp}}$ provides unbiased measurements with known Gaussian uncertainty $\sigma^2_{\text{exp}}(s)$ at states $s \in \mathcal{S}_{\text{exp}} \subset \mathcal{S}$. This is the standard treatment of experimental data in aerodynamics [**?**].

2.4 The Quality Certification Problem

Given a geometry Γ, a set of M experts $\{f_1, \dots, f_M\}$, and ground-truth anchors at states $\mathcal{S}_{\text{exp}}$, the SCX quality certification problem is:

(i) **Detection**: Can we bound the probability of missing a systematic error in any expert?

(ii) **Attribution**: When experts disagree, can we identify which expert(s) are in error?

(iii) **Ranking**: Can we quantitatively rank experts by quality on known benchmarks?

(iv) **Extrapolation**: What quality guarantees transfer to states $s \notin \mathcal{S}_{\text{exp}}$?

3 Theorem 1: SCX Noise Detection for CFD

We now prove the core noise detection theorem that bounds the probability of missing systematic errors when M solvers are applied to the same aerodynamic geometry.

3.1 Setup

Consider M experts $\{f_1, \ldots, f_M\}$ applied to a geometry Γ at K distinct flow states $\{s_1, \ldots, s_K\}$. Let $d_{ij}^{(k)} = d_{\mathcal{O}}(f_i(s_k), f_j(s_k))$ be the pairwise discrepancy between experts i and j at state s_k.

Definition 3.1 (Systematic Error)**.** Expert i has a *systematic error* of magnitude $\delta > 0$ at state s relative to ground truth $f^*(s)$ if:

$$d_{\mathcal{O}}(f_i(s), f^*(s)) > \delta \tag{5}$$

[A4] (Pairwise Detectability): If an expert i has systematic error $> \delta$ at state s, then for at least fraction $\gamma \in (0, 1]$ of other experts $j \neq i$, we have $d_{ij}(s) > \delta/2$.

Justification: If f_i deviates from ground truth by more than δ, and most other experts are within $\delta/2$ of ground truth (i.e., they are "good"), then by triangle inequality $d_{ij} \geq d(f_i, f^*) - d(f_j, f^*) > \delta - \delta/2 = \delta/2$. This assumption formalizes the intuition that a bad solution looks different from most good solutions.

3.2 Theorem Statement

Theorem 3.2 (SCX Noise Detection Bound for CFD)**.** *[rigor: full]Let $M \geq 3$ experts be applied to K independent flow states. Assume **[A4]** holds with detectability fraction γ. Define the consensus indicator for expert i at state k:*

$$C_i^{(k)} = \mathbf{1}\left(\frac{1}{M-1}\sum_{j\neq i} \mathbf{1}(d_{ij}^{(k)} \leq \delta/2) \geq 1 - \theta\right) \tag{6}$$

where $\theta \in (0, 1 - \gamma]$ is a consensus threshold. Let E_i be the event that expert i has systematic error $> \delta$ at a randomly chosen state but is not detected (i.e., $C_i = 1$ even though error $> \delta$). Then:

$$\mathbb{P}(E_i) \leq \exp\left(-2(M-1)\,(\gamma - \theta)^2\right) \tag{7}$$

Proof. Fix a state s_k where expert i has systematic error $> \delta$. By **[A4]**, for at least $\gamma(M-1)$ other experts j, we have $d_{ij}^{(k)} > \delta/2$. Let $X_j = \mathbf{1}(d_{ij}^{(k)} \leq \delta/2)$ for $j \neq i$. Then $\mathbb{E}[X_j] \geq 1 - \gamma$ for the "good" experts (those within $\delta/2$ of ground truth), but more importantly, the fraction of experts with $X_j = 1$ among the $M-1$ other experts is at most $1 - \gamma$.

For the consensus to falsely indicate no systematic error, we need at least $(1-\theta)(M-1)$ of the $M-1$ other experts to satisfy $d_{ij} \leq \delta/2$, i.e., $X_j = 1$. Let $\bar{X} = \frac{1}{M-1}\sum_{j\neq i} X_j$. The false consensus event is $\bar{X} \geq 1 - \theta$.

Under the null that expert i has systematic error, the expected fraction of close experts is $\mathbb{E}[\bar{X}] \leq 1 - \gamma$. By Hoeffding's inequality for $M-1$ bounded random variables $X_j \in [0, 1]$:

$$\mathbb{P}\left(\bar{X} - \mathbb{E}[\bar{X}] \geq \gamma - \theta\right) \leq \exp\left(-2(M-1)(\gamma - \theta)^2\right) \tag{8}$$

Since $\mathbb{E}[\bar{X}] \leq 1 - \gamma$, we have $\bar{X} \geq 1 - \theta \implies \bar{X} - \mathbb{E}[\bar{X}] \geq (1-\theta) - (1-\gamma) = \gamma - \theta$, which gives the bound in (**??**). ☐

Remark 3.3 (Interpretation for Neural CFD). The bound in Theorem **??** is exponentially decreasing in M and $(\gamma - \theta)^2$. For neural CFD:

(i) **Many traditional experts** $\Rightarrow$ large M $\Rightarrow$ strong detection.

(ii) **Training data contamination** manifests as neural expert i agreeing too well with training-distribution states but diverging on out-of-distribution states—this becomes detectable when traditional experts disagree with the neural prediction.

(iii) **Mesh dependency** appears as systematic error that varies with mesh resolution; multiple mesh resolutions provide the necessary expert diversity.

(iv) **Turbulence model mismatch** surfaces when a learned closure disagrees with wind tunnel data but traditional RANS with different closures bracket the experimental result.

[L1] The Hoeffding bound assumes independence between expert errors, which is an idealization. In practice, experts share physical assumptions and may exhibit correlated errors. This can be partially addressed using an effective sample size $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$ where $\bar{\rho}$ is average pairwise error correlation, estimable via bootstrap [**?**]. The exponential convergence rate in M is preserved with a modified constant.

Corollary 3.4 (Joint Detection Across States)**.** *[rigor: partial]For K independent flow states, the probability of failing to detect a systematically erroneous expert at any state is bounded by:*

$$\mathbb{P}(\textit{miss at any } s_k) \leq K \exp\left(-2(M-1)(\gamma-\theta)^2\right) \tag{9}$$

This follows from the union bound over K states. For $K = 10$ typical benchmark states, $M = 8$ experts, $\gamma = 0.5$, $\theta = 0.2$: $\mathbb{P}(\text{miss}) \leq 10 \cdot e^{-2 \cdot 7 \cdot 0.09} = 10 \cdot e^{-1.26} \approx 10 \cdot 0.284 = 2.84$, which exceeds 1—the union bound is too loose for practical use. A tighter bound via Slepian's inequality or Monte Carlo calibration is needed in practice.

[L2] The union bound in Corollary **??** is conservative and becomes vacuous for moderate K and M. In practice, we recommend Bonferroni-Holm correction or permutation testing on the actual expert discrepancy matrix.

4 Theorem 2: Unidentifiability of Error Sources in Neural CFD

When a neural CFD solver disagrees with wind tunnel data, practitioners and reviewers naturally ask: *what caused the error?* This theorem establishes a fundamental limit: without declared assumptions, the error source is unidentifiable.

4.1 Setup

Consider a neural CFD solver f_{NN} with learnable parameters $\boldsymbol{\theta}$ trained on dataset $\mathcal{D}$. The error $d_{\mathcal{O}}(f_{\text{NN}}(s), f^*(s))$ at test state $s \notin \mathcal{D}$ can arise from multiple distinct sources:

(i) **Architectural error** ϵ_{arch}: The neural network class cannot represent the true solution operator with any parameter setting.

(ii) **Training data insufficiency** ϵ_{data}: The training set $\mathcal{D}$ does not adequately cover the test flow regime.

(iii) **Optimization error** ϵ_{opt}: Training converged to a local minimum rather than the global optimum.

(iv) **Turbulence model error** ϵ_{turb}: Even the ground-truth physical model (e.g., RANS equations) is an approximation.

(v) **Numerical discretization error** ϵ_{disc}: The mesh or time step used in training differs from the reference.

Let $\mathcal{E} = \{\epsilon_{\text{arch}}, \epsilon_{\text{data}}, \epsilon_{\text{opt}}, \epsilon_{\text{turb}}, \epsilon_{\text{disc}}\}$ be the set of possible error sources.

4.2 Theorem Statement

Theorem 4.1 (Unidentifiability of Neural CFD Error Sources). *[rigor: full]Let f_{NN} be a neural CFD solver with observed test error $\Delta = d_{\mathcal{O}}(f_{NN}(s), f^*(s)) > 0$. Without declared assumptions specifying which error sources are bounded, there exist at least two distinct partitions of Δ among $\mathcal{E}$ that are observationally equivalent—i.e., produce the same observed output $f_{NN}(s)$ but attribute the error to different causes.*

Proof. We construct two distinct error source attributions that produce identical observed behavior.

Attribution A (Architecture blame): Assume that $\epsilon_{\text{data}} = \epsilon_{\text{opt}} = \epsilon_{\text{turb}} = \epsilon_{\text{disc}} = 0$, and the entire error Δ is due to ϵ_{arch}:

$$f_{\text{NN}}^{(A)}(s) = f_{\text{true}}(s) + \epsilon_{\text{arch}}, \quad \|\epsilon_{\text{arch}}\| = \Delta \tag{10}$$

Attribution B (Data insufficiency blame): Assume $\epsilon_{\text{arch}} = \epsilon_{\text{opt}} = \epsilon_{\text{turb}} = \epsilon_{\text{disc}} = 0$, but ϵ_{data} accounts for the error. However, training data insufficiency means the network parameters learned from $\mathcal{D}$ differ from the optimal parameters that would be learned from a sufficiently rich dataset $\mathcal{D}'$:

$$f_{\text{NN}}^{(B)}(s) = f_{\text{NN}}(s; \boldsymbol{\theta}(\mathcal{D}')) + \epsilon_{\text{data}}, \quad \|\epsilon_{\text{data}}\| = \Delta \tag{11}$$

Observational equivalence: Both attributions yield the same predicted output $f_{\text{NN}}(s) = f_{\text{NN}}^{(A)}(s) = f_{\text{NN}}^{(B)}(s)$ and the same test error Δ. Without further assumptions—such as:

- **[AA1]** A known upper bound on $\|\epsilon_{\text{arch}}\|$ from universal approximation analysis,
- **[AA2]** A known upper bound on $\|\epsilon_{\text{data}}\|$ from dataset coverage analysis,
- **[AA3]** A known upper bound on $\|\epsilon_{\text{turb}}\|$ from turbulence model validation,

—the attributions are indistinguishable from output observations alone.

Generalization to $|\mathcal{E}|$ sources: Let $\mathbf{e} = (e_1, \ldots, e_5)^T$ be the vector of error source magnitudes. Any vector $\mathbf{e}$ satisfying $\sum_i e_i = \Delta$ and the constraint that changing which source is "responsible" does not change the prediction is observationally equivalent. The set of such vectors is a 4-dimensional simplex, and without additional constraints (assumptions), any point on this simplex is a valid attribution. □

Remark 4.2 (Practical Implication for Neural CFD Papers). Theorem **??** has a direct consequence for neural CFD research: every paper claiming a neural method "improves" CFD must explicitly declare which error sources it addresses and which it assumes bounded. Without such declarations, a reviewer cannot determine whether a reported improvement is due to the claimed innovation or to favorable training data, better optimization, or a different discretization.

[A5] (Required Declarations for Neural CFD Papers): For any neural CFD contribution, the following must be declared:

(i) Training data coverage: which flow regimes are in $\mathcal{D}$ and which are test-only.

(ii) Architecture capacity: explicit statement of known approximation error bounds, or their absence.

(iii) Turbulence model baseline: which physical model underlies the training data, and its known limitations.

(iv) Optimization guarantee: whether training reached a local or global optimum, and how this was assessed.

(v) Mesh/convergence: mesh resolution used in training, and tests of prediction quality at different resolutions.

[L3] The construction in Theorem **??** uses two extreme attributions. Realistic scenarios involve partial contributions from multiple sources, and the theorem establishes a *lower bound* on unidentifiability: even in the simplest case, error source attribution requires assumptions. In practice, the ambiguity is larger.

5 Theorem 3: Situs Encoding Guarantee for Aerodynamic Geometry

The Situs encoding [?] maps physical objects into a metric space that respects geometric symmetries. For aerodynamic applications, we need guarantees that the encoding preserves the information relevant to flow prediction.

5.1 Situs Encoding for Airfoils and Wings

Definition 5.1 (Situs Encoding of an Airfoil). Let $\Gamma \subset \mathbb{R}^2$ be an airfoil profile defined by N coordinate points $\{(\xi_i, \zeta_i)\}_{i=1}^N$ in normalized chord coordinates. The Situs encoding $\Phi : \Gamma \mapsto \mathbf{z} \in \mathcal{Z} \subset \mathbb{R}^{d_z}$ is:

$$\Phi(\Gamma) = \left[\tilde{a}_0, \tilde{b}_0, \tilde{a}_1, \tilde{b}_1, \ldots, \tilde{a}_K, \tilde{b}_K\right] \tag{12}$$

where $\{\tilde{a}_k, \tilde{b}_k\}_{k=0}^K$ are the amplitude-normalized and translation-invariant Fourier coefficients of the camber line and thickness distribution, defined as:

$$\tilde{a}_k = \frac{a_k}{\max_j |a_j| + \epsilon}, \quad \tilde{b}_k = \frac{b_k}{\max_j |b_j| + \epsilon} \tag{13}$$

$$a_k = \frac{1}{\pi}\int_0^{2\pi} \zeta_{\text{camber}}(\theta)\cos(k\theta)d\theta, \quad b_k = \frac{1}{\pi}\int_0^{2\pi} \zeta_{\text{camber}}(\theta)\sin(k\theta)d\theta \tag{14}$$

where $\zeta_{\text{camber}}(\theta) = (\zeta_{\text{upper}}(\theta) + \zeta_{\text{lower}}(\theta))/2$ paramaterized by $\theta \in [0, 2\pi]$.

[A6] (Smooth Geometry): The airfoil profile Γ is C^2-continuous (twice differentiable). This holds for all standard NACA airfoils and most industrial designs. Discontinuities (e.g., sharp trailing edges) are handled by Fourier approximation with controlled truncation error.

[A7] (Flow Sensitivity to Geometry): The aerodynamic coefficients C_L, C_D, C_M are Lipschitz continuous in the Situs encoding of the geometry. Specifically, there exists $L_g > 0$ such that for any two geometries Γ_1, Γ_2:

$$d_{\mathcal{O}}(f(\Gamma_1), f(\Gamma_2)) \leq L_g \cdot \|\Phi(\Gamma_1) - \Phi(\Gamma_2)\|_2 \tag{15}$$

for the same flow conditions (M_∞, Re, α) and the same expert solver f.

Justification: This is a regularity assumption on the Navier-Stokes solution operator. For subsonic attached flows, smooth geometry perturbations produce smooth changes in pressure distribution [?]. For transonic flows with shocks, the Lipschitz constant L_g may be large but finite since shock location varies continuously with geometry [?].

Theorem 5.2 (Situs Encoding Guarantee for Aerodynamics). *[rigor: full]Let Γ be an airfoil profile satisfying **[A6]** and **[A7]**. Let $\hat{\Gamma}_K$ be the K-term Situs reconstruction:*

$$\hat{\Gamma}_K = \Phi^{-1}\left([\tilde{a}_0, \tilde{b}_0, \ldots, \tilde{a}_K, \tilde{b}_K, 0, \ldots]\right) \tag{16}$$

where coefficients beyond K are zeroed. Then for any C^2 airfoil, the truncation error in the Situs space satisfies:

$$\left\|\Phi(\Gamma) - \Phi(\hat{\Gamma}_K)\right\|_2 \leq \frac{C_\Gamma}{K^{3/2}} \tag{17}$$

where C_Γ depends on the maximum curvature and camber of Γ. Consequently, the aerodynamic prediction error due to geometry truncation is bounded by:

$$d_{\mathcal{O}}(f(\Gamma), f(\hat{\Gamma}_K)) \leq L_g \cdot \frac{C_\Gamma}{K^{3/2}} \tag{18}$$

Proof. For a C^2 periodic function (the camber line after parameterization), the Fourier coefficients decay as $|a_k|, |b_k| = O(k^{-2})$ [?]. Specifically, there exists C_0 such that $|a_k|, |b_k| \leq C_0/k^2$ for all $k \geq 1$.

The Situs encoding normalizes by $\max_j |a_j| + \epsilon$, so the normalized coefficients decay similarly:

$$|\tilde{a}_k|, |\tilde{b}_k| \leq \frac{C_0}{(\max_j |a_j| + \epsilon)k^2} \tag{19}$$

The L^2 norm of the truncated tail is:

$$\left\| \Phi(\Gamma) - \Phi(\hat{\Gamma}_K) \right\|_2^2 = \sum_{k=K+1}^{\infty} (\tilde{a}_k^2 + \tilde{b}_k^2) \leq 2 \sum_{k=K+1}^{\infty} \left(\frac{C_0}{C_m k^2} \right)^2 \tag{20}$$

where $C_m = \max_j |a_j| + \epsilon$.

For large K, $\sum_{k=K+1}^{\infty} k^{-4} \leq \int_K^\infty x^{-4} dx = \frac{1}{3K^3}$. Thus:

$$\left\| \Phi(\Gamma) - \Phi(\hat{\Gamma}_K) \right\|_2 \leq \sqrt{\frac{2C_0^2}{C_m^2}} \cdot \frac{1}{\sqrt{3}K^{3/2}} = \frac{C_\Gamma}{K^{3/2}} \tag{21}$$

with $C_\Gamma = \sqrt{2/3} \cdot C_0/C_m$.

The aerodynamic bound follows directly from **[A7]**. □

Remark 5.3 (Symmetry Preservation). The Situs encoding preserves three geometric symmetries crucial for aerodynamics:

(i) **Reflection:** Camber sign reversal corresponds to flipping $\{\tilde{a}_k\}$ signs, which preserves the L^2 norm.

(ii) **Scaling:** Amplitude normalization $(\max_j |a_j| + \epsilon)^{-1}$ makes the encoding scale-invariant.

(iii) **Translation:** The zero-frequency term a_0 corresponds to mean camber offset; its normalization eliminates translation sensitivity.

[L4] The bound $O(K^{-3/2})$ assumes C^2 geometry. For airfoils with sharp curvature changes (e.g., highly cambered supercritical airfoils), the effective smoothness order may be lower, yielding slower convergence. The constant C_Γ must be estimated per geometry class.

[L5] The Lipschitz constant L_g in **[A7]** is not known a priori and must be estimated empirically via finite differences on a calibration set of geometries. We provide an estimation procedure in the experimental protocol (§8).

6 Cercis Score for Aerodynamic Simulation

The Cercis score [?] quantifies the quality of an expert through two axes: **Q** (accuracy on benchmarks) and **N** (novelty of coverage). We adapt this to aerodynamic CFD.

6.1 Accuracy Score Q

Let $\mathcal{B} = \{b_1, \ldots, b_{|\mathcal{B}|}\}$ be a set of standard aerodynamic benchmark cases. For each benchmark $b = (M_\infty, Re, \Gamma)$, we have reference data (wind tunnel or highly resolved CFD) for $C_L, C_D, C_M, \mathbf{C}_p$.

Definition 6.1 (Accuracy Score)**.** For expert f_i, the accuracy score on benchmark set $\mathcal{B}$ is:

$$Q(f_i; \mathcal{B}) = 1 - \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \min\left(1, \frac{E_i(b)}{\tau_b}\right) \tag{22}$$

where:

- $E_i(b) = d_{\mathcal{O}}(f_i(s_b), f^*(s_b))$ is the error on benchmark b.
- τ_b is a benchmark-specific tolerance (see Table **??**).

$Q \in [0, 1]$, with $Q = 1$ indicating errors below tolerance on all benchmarks.

Table 1: Benchmark tolerances for Cercis Q-score computation. Tolerances are based on typical experimental uncertainty and engineering acceptability thresholds [**?**].

Benchmark	τ_{C_L}	τ_{C_D} (counts)	τ_{C_M}	τ_{C_p}
NACA 0012 (subsonic)	0.02	5	0.005	0.05
NACA 4412 (cambered)	0.03	8	0.008	0.06
ONERA M6 wing	0.05	10	0.010	0.10
NASA CRM (transonic)	0.05	10	0.010	0.10
DLR-F6 (wing-body)	0.05	15	0.010	0.12
NACA 0012 (transonic)	0.05	15	0.010	0.12

6.2 Novelty Score N

The novelty score measures how extensively a method has been validated across flow regimes.

Definition 6.2 (Novelty Score)**.** For expert f_i with published validation states $\mathcal{S}_i^{\text{pub}} \subset \mathcal{S}$:

$$N(f_i) = \frac{|\mathcal{S}_i^{\text{pub}} \setminus \mathcal{S}_{\text{ref}}|}{|\mathcal{S}_{\text{target}}|} \tag{23}$$

where $\mathcal{S}_{\text{ref}}$ is the set of flow states for which traditional CFD is considered reliable (e.g., attached subsonic flows with $Re \leq 10^7$), and $\mathcal{S}_{\text{target}}$ is the target coverage of interest (e.g., full flight envelope including transonic buffet and high-lift conditions).

6.3 Cercis Score

Definition 6.3 (Cercis Score)**.**

$$\mathcal{C}(f_i) = Q(f_i; \mathcal{B}) \cdot N(f_i) \tag{24}$$

The multiplicative form penalizes methods that achieve high accuracy only on easy regimes (N small) and methods that claim broad applicability without demonstrated accuracy (Q small).

Table 2: Estimated Cercis scores for published neural CFD methods. ✓ = reported with error below tolerance; ∼ = reported but above tolerance; blank = not reported.

Method	Benchmark Coverage				Q	N	$\mathcal{C}$
	0012	4412	M6	CRM			
FNO Airfoil [?]	✓	∼			0.35	0.15	0.053
MeshGraphNet [?]	✓	✓			0.40	0.20	0.080
DeepTurb [?]	✓	∼			0.30	0.10	0.030
PINN-NSFnet [?]	∼				0.15	0.05	0.008
GNN-Aero [?]	✓	✓	✓	✓	0.65	0.40	0.260
SA-RANS baseline	✓	✓	✓	✓	0.70	0.50	0.350
k-ω SST baseline	✓	✓	✓	✓	0.78	0.50	0.390

6.4 Ranking of Published Neural CFD Methods

Table **??** presents estimated Cercis scores for published neural CFD methods, based on reported results. Where a method has not reported results on a benchmark, we assign $E_i(b) = \tau_b$ (i.e., error equals tolerance threshold, yielding zero partial credit). **[L6]** Scores are approximate due to inconsistent reporting across papers; we urge authors to adopt standardized reporting.
The traditional RANS baselines achieve higher Cercis scores due to broader validation across benchmarks, even though neural methods may achieve higher accuracy on specific cases. This illustrates the SCX philosophy: **quality is not just accuracy on cherry-picked cases, but demonstrated reliability across the operational envelope**.

[L7] The Cercis ranking in Table **??** uses approximate values extracted from published figures and text. A definitive ranking requires recomputing all methods on the same benchmark suite with consistent metrics, which we propose as a community effort in §8.

7 Multi-Expert Architecture with Yajie Consensus

We now describe a practical architecture for SCX-based quality auditing of CFD predictions.

7.1 Expert Ensemble

The multi-expert system comprises:

E1: E1: k-ω SST RANS (widely validated turbulence model).

E2: E2: k-ε RANS (alternative closure, known to differ in adverse pressure gradients).

E3: E3: Spalart-Allmaras RANS (efficient one-equation model).

E4: E4: LES (scale-resolving, computationally expensive but high fidelity).

E5: E5: FNO-based neural operator (speed-oriented).

E6: E6: GNN-based mesh solver (geometry-flexible).

E7: E7: PINN with RANS loss (physics-constrained).

E8: E8: Wind tunnel data (ground-truth anchor, available only at $\mathcal{S}_{\text{exp}}$).

7.2 Yajie Consensus Mechanism

The Yajie consensus mechanism [?] processes expert outputs through two stages:

Stage 1: Discrepancy Matrix. For each state s_k, compute the $M \times M$ pairwise discrepancy matrix $\mathbf{D}^{(k)}$ with entries $D_{ij}^{(k)} = d_{\mathcal{O}}(f_i(s_k), f_j(s_k))$.

Stage 2: Consensus Score. For each expert i, compute the consensus score:

$$S_i^{(k)} = \frac{1}{M-1} \sum_{j \neq i} w_{ij} \cdot \kappa(D_{ij}^{(k)}) \tag{25}$$

where w_{ij} are expert-pair weights reflecting known reliability differences, and $\kappa(d) = \exp(-d^2/\sigma^2)$ is a Gaussian kernel with bandwidth σ set to the median discrepancy.

Stage 3: Anomaly Flagging. Flag expert i at state k if $S_i^{(k)} < \tau_{\text{consensus}}$, where $\tau_{\text{consensus}} = 0.5$ by default.

[A8] (Yajie Anchoring): The wind tunnel expert (E8) receives weight $w_{8j} = \lambda \cdot w_{\text{base}}$ with $\lambda \geq 2$ when its measurement uncertainty $\sigma_{\text{exp}}(s_k)$ is below a threshold. This reflects the privileged status of experimental data while acknowledging its uncertainty.

7.3 Workflow

Algorithm 1 SCX Audit for a New Aerodynamic Simulation

Input: Geometry Γ, flow states $\{s_1, \ldots, s_K\}$, experts $\{f_1, \ldots, f_M\}$
Output: Quality report with anomaly flags, consensus scores, Cercis estimate
for $k = 1$ to K **do**
  for $i = 1$ to M **do**
    Compute $f_i(s_k)$ — expert prediction
  end for
  Compute discrepancy matrix $\mathbf{D}^{(k)}$
  Compute Yajie consensus scores $\{S_i^{(k)}\}$
  Flag anomalous experts (Theorem **??** check)
end for
For states $s_k \in \mathcal{S}_{\text{exp}}$ with wind tunnel data:
  Compute per-expert errors, identify systematic errors
  Apply Theorem **??** — note which error sources are unidentifiable
Compute Cercis score $\mathcal{C}(f_i)$ for each non-anchor expert
return Quality report

[L8] The multi-expert architecture requires running all M experts for each state, which is computationally expensive. However, (a) many traditional CFD solutions can be precomputed and cached for standard geometries, (b) neural experts are fast at inference time ($<$ 1 second per state), and (c) the audit is designed for certification workflows where computational cost is secondary to reliability.

8 Experimental Protocol

We propose a standardized experimental protocol for evaluating neural CFD methods under SCX audit. This protocol is designed for reproducibility and fair comparison.

8.1 Benchmark Suite

8.2 Mesh Convergence Protocol

To assess mesh dependency (a key concern for neural solvers), we define four mesh levels:

Table 3: Standard aerodynamic benchmarks for SCX CFD audit.

Benchmark	M_∞	Re (chord)	α range	Reference data
NACA 0012	0.15	6×10^6	0°–15°	[?]
NACA 0012	0.70	9×10^6	0°–4°	[?]
NACA 4412	0.15	6×10^6	0°–18°	[?]
ONERA M6	0.84	11.7×10^6	3.06°–6.06°	[?]
NASA CRM	0.85	5×10^6	2°–4°	[?]
DLR-F6	0.75	3×10^6	0°–2°	[?]
30P30N (high-lift)	0.20	9×10^6	4°–20°	[?]

(i) **Coarse:** $\sim 10^4$ cells (typical of neural CFD training).

(ii) **Medium:** $\sim 10^5$ cells (standard engineering RANS).

(iii) **Fine:** $\sim 10^6$ cells (mesh-converged RANS).

(iv) **Ultra-fine:** $\sim 10^7$ cells (LES/DNS resolution).

The mesh convergence index (MCI) is computed via Richardson extrapolation [?]:

$$\text{MCI} = \frac{|f_{\text{fine}} - f_{\text{medium}}|}{r^p - 1} \tag{26}$$

where r is the refinement ratio and p is the observed order of accuracy. Neural solvers must demonstrate that their prediction error does not exceed the MCI of traditional solvers at equivalent mesh resolution.

8.3 Neural Architecture Ablations

[A9] (Ablation Protocol): For each neural CFD method, report results under the following ablations:

(i) **Training set size**: 25%, 50%, 75%, 100% of available training data.

(ii) **Training flow regime**: train on subsonic only, test on transonic (and vice versa).

(iii) **Architecture depth**: vary number of layers/parameters by factors of 2× and 0.5×.

(iv) **With and without physics loss**: PINNs with and without the PDE residual term.

(v) **Mesh resolution mismatch**: train on one mesh level, test on another.

8.4 Turbulence Model Variation

For traditional CFD baselines, run the following turbulence model variants:

- k-ω SST (Menter, 1994)
- k-ε realizable (Shih et al., 1995)
- Spalart-Allmaras (Spalart & Allmaras, 1992)
- Transition SST (γ-Re_θ)
- Scale-Adaptive Simulation (SAS)

8.5 Evaluation Metrics

(i) **Integral quantities:** $\epsilon_{C_L} = |C_L^{\text{pred}} - C_L^{\text{ref}}|$, similarly for C_D, C_M.

(ii) **Pressure distributions:** L^2 error $\left\|\mathbf{C}_p^{\text{pred}} - \mathbf{C}_p^{\text{ref}}\right\|_{L^2(\Gamma_w)} / \left\|\mathbf{C}_p^{\text{ref}}\right\|_{L^2(\Gamma_w)}$.

(iii) **Shock location:** For transonic cases, $\Delta x_{\text{shock}}/c$, the chord-normalized difference in shock position.

(iv) **Separation point:** For high-angle cases, $\Delta x_{\text{sep}}/c$, the chord-normalized difference in separation location.

(v) **Consensus score:** Yajie $S_i^{(k)}$ as defined in (**??**).

(vi) **Cercis score:** $\mathcal{C}(f_i) = Q(f_i) \cdot N(f_i)$.

8.6 Assumption Declaration Template

Following Theorem **??**, every neural CFD submission must include:

Assumption Declaration (SCX-CFD v1.0)

A1. Data Coverage: [Flow regimes in training; regimes reserved for testing] **A2. Architecture Capacity:** [Known bounds or absence thereof] **A3. Physical Model:** [Underlying equations; known limitations of these equations] **A4. Optimization:** [Convergence criterion; evidence of local vs global optimum] **A5. Discretization:** [Training mesh resolution; tests at other resolutions] **A6. Reproducibility:** [Code availability; random seeds; hardware used]

9 Discussion

9.1 Relationship to Neural Operator Theory

Neural operators [**?**] provide a mathematical framework for learning mappings between infinite-dimensional function spaces, which is directly applicable to CFD where solutions are functions of space and time. Theorem **??** complements neural operator theory by providing a *detection* guarantee: while neural operators can theoretically approximate the Navier-Stokes solution operator (universal approximation in appropriate Sobolev spaces), the SCX framework detects when a trained operator *fails* to do so in practice.

The key connection is that neural operator convergence rates ($O(N^{-\alpha})$ where N is training samples and α depends on smoothness) provide an upper bound on ϵ_{arch} in Theorem **??**, partially resolving one source of unidentifiability when explicit convergence analysis is available.

9.2 The Turbulence Modeling Closure Problem

Turbulence modeling remains the central challenge in CFD [**?**]. The closure problem—that the Reynolds stresses $\overline{u_i'u_j'}$ introduce more unknowns than equations—has no closed-form solution. All turbulence models, whether traditional algebraic closures or learned neural closures, are approximations.

The SCX framework treats this honestly: **[A3]** in the RANS equations is explicitly the turbulence model assumption, and Theorem **??** clarifies that when a neural closure disagrees with wind tunnel data, the error may originate in the RANS framework itself rather than in the neural implementation. This prevents the common pitfall of "blaming the neural network" when the underlying physical model is the limiting factor.

9.3 Path to Certified Neural CFD for Industrial Use

Industrial adoption of neural CFD requires certification—evidence that predictions are reliable for safety-critical design decisions. The SCX framework contributes three components toward this goal:

(i) **Detection** (Theorem **??**): Automated flagging of anomalous predictions when multiple experts disagree.

(ii) **Delineation** (Theorem **??**): Explicit boundaries of what can and cannot be concluded from disagreement.

(iii) **Benchmarking** (Cercis score, §6): Standardized, reproducible comparison across methods.

We do *not* claim that SCX provides certification by itself. Certification additionally requires:

- Formal verification of solver correctness (analogous to the Lax equivalence theorem for traditional methods).
- Bounds on extrapolation error to unseen geometries and flow conditions.
- Regulatory acceptance (e.g., EASA/FAA certification processes).

9.4 Limitations

We explicitly acknowledge the following limitations of the SCX CFD audit framework:

[L4] **(Computational cost)** Running $M \geq 8$ experts on K benchmark cases requires significant computational resources, though amortized over a certification process this is acceptable.

[L5] **(Correlated expert errors)** Theorem **??** assumes independent errors. Traditional RANS models share physical assumptions and produce correlated errors, particularly in separated flows where all RANS closures fail similarly. The effective M_{eff} correction partially addresses this but requires empirical correlation estimation.

[L6] **(Wind tunnel coverage)** Ground-truth anchors exist only at specific M_∞, Re, α combinations. Extrapolation to flight conditions requires additional assumptions beyond the scope of this paper.

[L7] **(RANS limitation)** All traditional experts based on RANS share the fundamental limitation of Reynolds averaging: unsteady phenomena (buffet, flutter, vortex shedding) cannot be captured. Neural closures trained on RANS data inherit this limitation.

[L8] **(Cercis score subjectivity)** The benchmark selection and tolerance thresholds in Table **??** embed engineering judgment. Different applications (e.g., wind turbine vs. transonic wing design) may require different benchmarks and tolerances.

[L9] **(Situs Lipschitz estimation)** The constant L_g in **[A7]** is problem-dependent. Our estimation via finite differences on calibration geometries provides a lower bound; the true L_g may be larger for rarely encountered geometries.

9.5 Future Work

Several directions emerge from this framework:

(i) **Rigorous Situs Lipschitz estimation:** Derive L_g analytically for simplified geometries (e.g., Joukowski airfoils) using conformal mapping and potential flow theory, then extend numerically to general geometries.

(ii) **Effective sample size calibration:** Develop a systematic procedure for estimating $\bar{\rho}$ in the correlated-expert setting through designed perturbation experiments.

(iii) **Neural CFD certification suite:** Implement the experimental protocol (§8) as an automated benchmark suite with standardized input/output formats, enabling reproducible comparison across methods.

(iv) **Extrapolation bounds:** Extend Theorem **??** to bound prediction quality at flow states $s \notin \mathcal{S}_{\text{exp}}$ using Lipschitz continuity of the solution operator in state space.

(v) **Integration with turbulence model development:** Apply the SCX audit during turbulence model development to provide continuous quality monitoring and prevent regression.

10 Conclusion

We have presented the SCX quality audit framework applied to neural computational fluid dynamics for finite element aerodynamics. Three theorems—noise detection (§3), error source unidentifiability (§4), and Situs encoding guarantee (§5)—provide formal foundations for quality assessment. The Cercis score (§6) enables quantitative comparison, and the multi-expert Yajie architecture (§7) provides a practical auditing workflow.

The framework does not claim that neural CFD is unreliable. Rather, it provides the tools to *determine* when and why it is reliable. By making assumptions explicit, errors detectable, and comparisons standardized, the SCX approach advances the path toward certified neural CFD for industrial aerodynamic design.

CoreProposition (Core Proposition): Traditional CFD provides implicit quality assurance through multi-expert redundancy (different turbulence models, grid resolutions, numerical schemes). Neural CFD methods lack such guarantees. The SCX audit framework provides the missing quality certification layer for neural CFD by explicitizing assumptions, detecting systematic errors, and standardizing benchmarks.

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A Assumption Map

Table **??** provides a complete map of all assumptions declared in this paper, their types, and their dependencies.

Table 4: Complete assumption map for SCX CFD audit.

ID	Description	Type	Depends on
[A0]	Neural CFD lacks formal quality guarantees	Empirical	Literature review
[A1]	Deterministic experts	Modeling	Numerical precision
[A2]	Compact output space	Modeling	Physical bounds
[A3]	Unbiased wind tunnel data	Empirical	Experimental protocol
[A4]	Pairwise detectability (γ)	Structural	Expert diversity
[A5]	Required declarations for NN-CFD	Normative	Theorem **??**
[A6]	C^2 airfoil geometry	Regularity	Geometry class
[A7]	Flow-geometry Lipschitz continuity	Physical	Navier-Stokes regularity
[A8]	Yajie anchoring weights	Methodological	Expert calibration
[A9]	Ablation protocol completeness	Methodological	Experimental design

B Detailed Proof of Theorem ?? with Correlation Correction

[rigor: partial]We provide a more detailed treatment of Theorem **??** accounting for expert error correlation.

Assume M experts with pairwise error correlation $\rho_{ij} = \mathrm{Corr}(X_i, X_j)$ where $X_i = \mathbf{1}(d(f_i, f^*) > \delta/2)$. The effective number of independent experts is [?]:

$$M_{\text{eff}} = \frac{M}{1 + (M-1)\bar{\rho}} \tag{27}$$

where $\bar{\rho} = \frac{2}{M(M-1)} \sum_{i<j} \rho_{ij}$ is the average pairwise correlation.

Replacing M with M_{eff} in the Hoeffding bound yields:

$$\mathbb{P}(E_i) \leq \exp\left(-2M_{\text{eff}}(\gamma - \theta)^2\right) \tag{28}$$

This bound is conservative when $\bar{\rho} > 0$ (the typical case for traditional CFD experts that share physical assumptions). A bootstrap procedure for estimating $\bar{\rho}$ from the discrepancy matrix $\mathbf{D}$ is:

(i) Resample K states with replacement B times.

(ii) For each bootstrap sample b, compute the vector of per-expert errors $\mathbf{e}^{(b)} \in \mathbb{R}^M$.

(iii) Estimate $\hat{\rho}_{ij} = \mathrm{Corr}(\mathbf{e}_i, \mathbf{e}_j)$ across bootstrap samples.

(iv) Compute $\bar{\rho}$ and M_{eff}.

[L13] This bootstrap estimate is itself noisy when K is small. For practical SCX audits with $K \geq 20$, bootstrap estimation provides reasonable accuracy.